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Atomic layer deposition of MoO_x thin films using $\text{Mo}(\text{PrCp})_2\text{H}_2$ and O_3

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ABSTRACT

This work studied the growth of molybdenum oxide thin films with thermal atomic layer deposition (ALD) using $\text{Mo}(\text{}^i\text{PrCp})_2\text{H}_2$ and O_3 as precursors. Growth parameters were determined by varying growth temperature and precursor dose times. ALD growth was exhibited in a temperature range of 100–200 °C. The growth per cycle ranged from 1.3 to 1.7 Å/cycle with a standard uniformity parameter of <5%. Attempts to grow films using H_2O as an oxygen source showed no significant growth. Film properties were measured using spectroscopic ellipsometry, x-ray reflectivity, x-ray diffraction, x-ray photoelectron spectroscopy, and infrared spectroscopy.

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I. INTRODUCTION

Atomic layer deposition (ALD) is a widely used subset of chemical vapor deposition that yields conformal films with precise thickness control.¹ Through alternating vapor phase precursor exposures, a series of self-limiting reactions take place on the surface of a substrate resulting in film growth. Films deposited using ALD can be grown on large substrate areas while maintaining high uniformity and conformality.¹ Because of these unique properties, ALD has been used to deposit a wide range of materials for semiconductors and other device and material applications.²

Within ALD materials, ultrathin transition metal oxides have seen a particularly wide interest and use ranging from dielectrics to semiconducting oxides. The compositional focus of this work, MoO_x ($x \sim 3$), has attracted interest for its electrical and catalytic properties.^{3,4} MoO_x also produced promising results as a hole-selective material for application in solar cells such as in silicon heterojunctions.^{5–7} MoO_3 has previously been grown through various methods including ALD, thermal evaporation, sputtering, solution casting, and CVD.^{8–12} Notable precursors for ALD of MoO_3 include molybdenum hexacarbonyl ($\text{Mo}(\text{CO})_6$)¹³ and Bis(*t*-butylimido)bis(dimethylamino)molybdenum(VI) [$(\text{}^t\text{BuN})_2(\text{NMe}_2)_2\text{Mo}$].⁶ $\text{Mo}(\text{CO})_6$ and ozone (O_3) have been used to grow substoichiometric molybdenum oxide ($x < 3$) thin films in a temperature range of 152–172 °C. A growth rate of 0.7 Å/cycle was found at a growth temperature of 167 °C. The low thermal

stability of $\text{Mo}(\text{CO})_6$ is shown by the narrow 10 °C thermal growth window leading to nonuniform growth and CVD-like behavior across a wide range of temperatures. $(\text{}^t\text{BuN})_2(\text{NMe}_2)_2\text{Mo}$ and O_3 have been reported to grow high purity MoO_x films at deposition temperatures of 100–300 °C, but with a low growth rate of 0.4 Å/cycle. X-ray photoelectron spectroscopy (XPS) on these films demonstrated low levels of carbon contamination. However, the somewhat narrow ALD temperature window of $\text{Mo}(\text{CO})_6$ and the low growth per cycle (GPC) of the amido-based Mo precursor motivate the investigation of new Mo precursors that will expand the temperature range and growth window.

In this work, bis-isopropyl cyclopentadienyl molybdenum dihydride [$\text{Mo}(\text{}^i\text{PrCp})_2\text{H}_2$] was investigated as a molybdenum precursor with O_3 and H_2O as oxygen precursors. The ALD temperature window and growth behavior was quantified along with the physical and chemical properties of the films. Films grown using this precursor were compared to films grown using $(\text{}^t\text{BuN})_2(\text{NMe}_2)_2\text{Mo}$ and attempts to grow films using $\text{Mo}(\text{CO})_6$ are also discussed.

II. EXPERIMENT

A. Film growth

MoO_x films were deposited using a Cambridge NanoTech Savannah S100 cross flow reactor with a nitrogen flow rate of

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20 SCCM. ALD using $\text{Mo}(\text{}^i\text{PrCp})_2\text{H}_2$ (Ereztech, LLC) and O_3 or H_2O was investigated by adjusting the Mo precursor temperature, deposition temperature, precursor pulse time, and oxidant pulse time to determine suitable conditions for growth and the ALD temperature window. O_3 was pulsed with a concentration of ~ 3 wt. % in O_2 through an orifice into the ALD chamber. Mo precursor and oxidant pulses were separated by continuous 20 SCCM nitrogen gas purge with purge times that were longer for lower substrate temperatures to allow adequate purging of residual gases and quantified below.

Growth using $(\text{}^t\text{BuN})_2(\text{NMe}_2)_2\text{Mo}$ and O_3 was performed based on literature description.⁶ The deposition temperature was set to 200 °C and $(\text{}^t\text{BuN})_2(\text{NMe}_2)_2\text{Mo}$ temperature was set to 50 °C. $(\text{}^t\text{BuN})_2(\text{NMe}_2)_2\text{Mo}$ was delivered to the chamber using a 0.10 s nitrogen boost, during which time the ALD precursor cylinder was pressurized with N_2 (g) to approximately 20 psi (gauge), followed by a 2 s wait and 0.75 s pulse into the chamber. This boost, wait, and pulse procedure was repeated for 15 sequential “microdoses” to achieve saturation for a given Mo precursor exposure step. O_3 was pulsed continuously through the orifice for 48 s. A purge time of 20 s between Mo and O_3 half reactions was used. For all ALD growths on silicon, depositions were performed on 0.5×0.5 -in. test-grade p-type 525 μm thick Si(100) coupons. Coupons were ultrasonically cleaned for 5 min in isopropyl alcohol, rinsed in 18.2 M Ω water, and treated with UV ozone for 5 min prior to loading into the chamber.

For an attempt to grow MoO_x films using $\text{Mo}(\text{CO})_6$, precursor pulse times were determined based on the literature.¹³ The Mo precursor and O_3 were pulsed for 4 and 40 s, respectively. The Mo precursor was held at 70 °C and deposition temperature was determined by varying the chamber temperature from 120 to 170 °C. No ALD growth was measured for any deposition using $\text{Mo}(\text{CO})_6$ as a Mo precursor.

Saturation studies for $\text{Mo}(\text{}^i\text{PrCp})_2\text{H}_2$ were performed at a chamber temperature of 150 °C with the Mo precursor held at 80 °C. As dosage saturation behavior was investigated for both co-reactants, the remaining parameters, including precursor dose times [$\text{Mo}(\text{}^i\text{PrCp})_2\text{H}_2$: 0.25 s and O_3 : 40 s], and the number of precursor pulses [$\text{Mo}(\text{}^i\text{PrCp})_2\text{H}_2$: 6 pulses and O_3 : 1 pulse] were kept constant. The N_2 boost mode was also used for the $\text{Mo}(\text{}^i\text{PrCp})_2\text{H}_2$ precursor. In a separate experiment, deposition temperature was varied from 100 to 300 °C in increments of 25 °C to determine the upper and lower limits of the ALD growth temperature window. Uniformity across the chamber was quantified using the standard uniformity parameter (SUP), which is the coefficient of variation in thickness across all measured samples from a given deposition,

$$\text{SUP} = \frac{\text{Standard Deviation of Thickness}}{\text{Average Thickness}}.$$

B. Film characterization

MoO_x was deposited using the saturated precursor doses at deposition temperatures of 100, 150, and 200 °C, using 240, 212, and 189 cycles, respectively, to grow a film thickness of ~ 30 nm at each growth temperature. Two samples from the 200 °C deposition

underwent an anneal at 350 °C for 20 min in air. MoO_x film thickness was measured using a J. A. Woollam VASE spectroscopic ellipsometer (SE) and data were fit with a Cauchy model. Extinction coefficient, refractive index, and thickness were found using SE. Thickness, density, and roughness were analyzed by x-ray reflectivity (XRR) using a Panalytical Empyrean x-ray diffractometer.¹⁴ Crystallinity was measured for as-deposited films on Si deposited at 100, 150, and 200 °C and a 350 °C-annealed film on Si deposited at 200 °C using the Panalytical Empyrean x-ray diffractometer for x-ray diffraction (XRD), measuring 2θ symmetric scans. Film composition and stoichiometry were measured using a SPECS XPS instrument. A survey scan was done to identify significant peaks, followed by high resolution core-level scans on the Mo 3d, O 1s, C 1s, and Si 2p peaks when applicable. Films were sputtered for 360 s with 5 keV Ar^+ to remove ~ 10 nm of material to provide an accurate measure of carbon content in the bulk of the film. Additionally, chemical species in the film were found using Fourier transform infrared (FTIR) spectroscopy. The FTIR spectra of the MoO_x films were collected in transmission mode in a Perkin Elmer 100 spectrometer equipped with N_2 gas purge. Double side polished float zone Si (100) wafers from Virginia Semiconductor with 2×6 cm² dimensions were used for FTIR. These chips were cleaned by sonication in IPA (isopropyl alcohol) and 18.2 M Ω water followed by 5 min UVO cleaning for 5 min prior to deposition. For background, the clean DSP wafer FTIR spectrum was collected prior to the deposition to isolate the signal from the MoO_x film. The measurements were averaged over 250 scans with 4 cm⁻¹ scan resolution. For composition comparison, MoO_x films of similar thickness were grown using $(\text{NtBu})_2(\text{NMe}_2)_2\text{Mo}$ ALD.

III. RESULTS AND DISCUSSION

Based on experimental vapor pressure measurements¹⁵ by Zorin *et al.*, heating the $\text{Mo}(\text{}^i\text{PrCp})_2\text{H}_2$ precursor to 80 °C was determined to be sufficient to provide adequate vapor pressure for ALD, while maintaining thermal stability. However, this condition alone did not result in uniform growth across the reactor chamber with standard vapor draw delivery. To increase the dosage of $\text{Mo}(\text{}^i\text{PrCp})_2\text{H}_2$, a 0.1 s boost of N_2 gas was introduced to the Mo precursor dosing step. Adding this boost increased uniformity and increased the GPC near the outlet relative to delivery based on purely vapor draw. The average GPC also increased significantly from 0.3 to 1.0 Å/cycle when using one pulse of Mo precursor without boost and with boost, respectively. For all remaining discussion, boost mode was used at a precursor temperature of 80 °C.

Growth saturation was investigated for both O_3 and $\text{Mo}(\text{}^i\text{PrCp})_2\text{H}_2$ independently by varying the dose of one precursor and keeping the other precursor dose constant (Fig. 1). Mo saturation was tested at a deposition temperature of 150 °C using a 40 s O_3 dose time. $\text{Mo}(\text{}^i\text{PrCp})_2\text{H}_2$ saturated at approximately six pulses, with a GPC of 1.5 Å/cycle. From four to eight Mo precursor microdoses, the SUP remained low (2.0%–4.2%) also indicating nearly ideal ALD behavior. Using these saturated Mo conditions (six Mo microdoses), O_3 dose time was separately varied. Saturation in the GPC occurred in the range of 20–40 s dose time with a GPC of 1.4–1.6 Å/cycle.

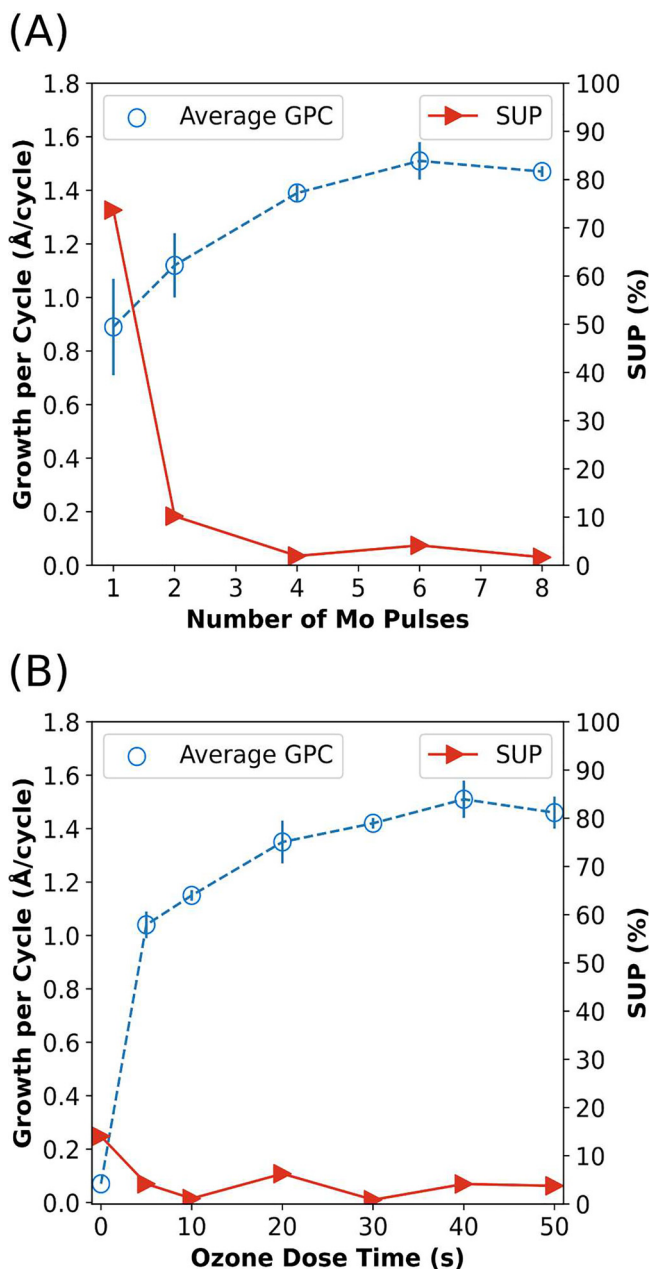


FIG. 1. Saturation curves for ozone (3 wt. %) and bis-isopropylcyclopentadienyl molybdenum dihydride. Sample growth temperatures were set to 150 °C. Pulsing sequences are (a) $\text{Mo}(\text{PrCp})_2\text{H}_2 + 20 \text{ s purge} + 40 \text{ s O}_3 + 20 \text{ s purge}$ and (b) $6 \times 0.25 \text{ s Mo}(\text{PrCp})_2\text{H}_2 + 20 \text{ s purge} + \text{O}_3 + 20 \text{ s purge}$.

Several other control experiments were conducted to test for reaction with water and thermal decomposition (CVD-like behavior) of the Mo precursor. $\text{Mo}(\text{PrCp})_2\text{H}_2$ was pulsed into the chamber without O_3 to test for decomposition at 150 °C and yielded a GPC of 0.1 Å/cycle over 50 cycles, indicating no

TABLE I. Varied purge times for various deposition temperatures.

Deposition temperature (°C)	Purge time (s)
100	30
125	25
150	20
175	17
200	10
250	8
300	4

significant CVD occurred at this temperature. H_2O was used instead of O_3 as an oxygen precursor to test its reactivity with $\text{Mo}(\text{PrCp})_2\text{H}_2$ and an average GPC of 0.1 Å/cycle was measured over 50 cycles. These data indicate that this Mo precursor is thermally stable at these temperatures and that H_2O is not sufficiently reactive to result in ALD growth (Table I).

The temperature dependence of the growth behavior of the $\text{Mo}(\text{PrCp})_2\text{H}_2/\text{O}_3$ pair indicates uniform growth ($\text{SUP} < 10\%$) from 100 to 200 °C (Fig. 2). The GPC increased from 100 to 300 °C from 1.26 to as high as 2.6 Å/cycle. When pulsed at deposition temperatures above 225 °C, the SUP increased beyond 10% and GPC indicates that $\text{Mo}(\text{PrCp})_2\text{H}_2$ may begin decomposing at these temperatures. A saturation study was not conducted at temperatures other than 150 °C; thus, the ALD window for this precursor based on the SUP appears to be from approximately 100 to 200 °C.

A comparison thermal study was performed using $\text{Mo}(\text{CO})_6$ and O_3 as precursors, with conditions similar to the ones used by

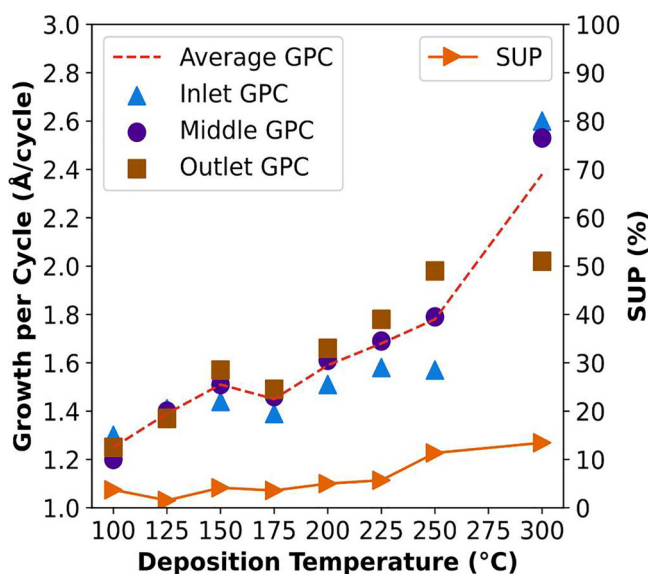


FIG. 2. Growth per cycle of MoO_x and SUP for various growth temperatures using the pulsing sequence of $6 \times 0.25 \text{ s Mo}(\text{PrCp})_2\text{H}_2 + \text{purge} + 40 \text{ s O}_3 + \text{purge}$. Purge time was varied with deposition temperature.

Diskus *et al.*¹³ Nonuniform growth was seen at the reported stable growth temperature of 167 °C. The average GPC was measured as 8.1 Å/cycle at 170 °C, dropping to 1.8 Å/cycle at 150 °C and ~0 Å/cycle at 140 °C in our experiments. Our results with the CO-based precursor are not consistent with prior results, with literature reporting the GPC at 0.75 Å/cycle at 167 °C. Our results are also indicative of the difficulty in achieving uniform and consistent ALD growth with that precursor.

Optical constants for ~30 nm thick films were obtained from SE (Table II). Values for refractive index (n) and extinction coefficient (k) are reported at a wavelength of 605 nm. Vos *et al.*¹⁶ found $n = 2.099$ when using $\text{Mo}(\text{NtBu})_2(\text{NMe}_2)_2$ and O_2 plasma as reactants. The value of n of crystalline $\alpha\text{-MoO}_3$ is reported to be 2.139.¹⁷ For the MoO_x deposited in this study at 100, 150, and 200 °C, the index of refraction was 1.44, 1.42, and 1.38. For comparison, films grown in our laboratory using $\text{Mo}(\text{NtBu})_2(\text{NMe}_2)_2$ had a refractive index of 1.62. Vos *et al.* found k to be 0.0164. k in crystalline $\alpha\text{-MoO}_3$ is measured as 0.006 31.¹⁸ Extinction coefficients were 0.000 31 and 0.0063 for deposition temperatures 100 and 150 °C, respectively. The 200 °C deposition was once again the outlier, as its extinction coefficient was raised to 0.15, significantly higher than the values for lower deposition temperatures. MoO_x deposited with $\text{Mo}(\text{NtBu})_2(\text{NMe}_2)_2$ in this study exhibited an extinction coefficient of 0.073. Generally, the low n values as compared to crystalline samples indicate low density material and possible contamination with residual ligand species.

Density and roughness values were obtained using XRR (Table II). The mass densities ranged from 2.39 to 2.85 g/cm³ and increased with increasing deposition temperature. These densities correspond to only 50%–60% of the density of $\alpha\text{-MoO}_3$ (4.78 g/cm³).¹⁹ The increase in density seen with the increase in deposition temperature is possibly due to the removal of organic contaminants within the film and greater adatom mobility. This trend in density disagrees with the trend in n value for the 200 °C sample. Refractive index is expected to increase monotonically with density, but instead has the opposite effect, as density seems to increase when n decreases at 200 °C. This discrepancy is possibly due to the significant presence of carbon found in XPS data (*vide infra*). Incorporation of carbon into the films can lead to low density values and changes to n . Films deposited with $\text{Mo}(\text{NtBu})_2(\text{NMe}_2)_2$ were found to have a density of 3.04 g/cm³. The larger density of films deposited using $\text{Mo}(\text{NtBu})_2(\text{NMe}_2)_2$ may be due to higher film purity. Literature reports the density of ALD MoO_x as 2.6 g/cm³ according to Pan *et al.*,⁹ which is lower than densities found when using $\text{Mo}(\text{PrCp})_2\text{H}_2$ except when depositing

TABLE II. Optical constants, density, and roughness for as-deposited MoO_x deposited with different precursors.

Mo precursor (deposition temperature)	n	k	Density (g/cm ³)	Roughness (nm)
$\text{Mo}(\text{NtBu})_2(\text{NMe}_2)_2$ (200 °C)	1.62	0.073	3.04	0.6
$\text{Mo}(\text{PrCp})_2\text{H}_2$ (100 °C)	1.44	0.000 31	2.39	0.7
$\text{Mo}(\text{PrCp})_2\text{H}_2$ (150 °C)	1.42	0.0063	2.62	1.6
$\text{Mo}(\text{PrCp})_2\text{H}_2$ (200 °C)	1.38	0.15	2.85	0.7

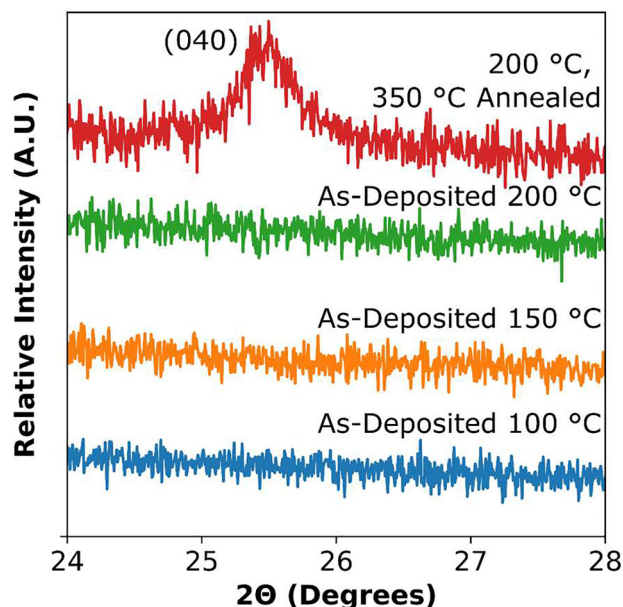


FIG. 3. X-ray diffraction pattern for as-deposited and annealed MoO_x growth with the $\text{Mo}(\text{iPrCp})_2\text{H}_2$ precursor for $2\theta = 24^\circ\text{--}28^\circ$. The annealed sample was heated to 350 °C in air for 20 min.

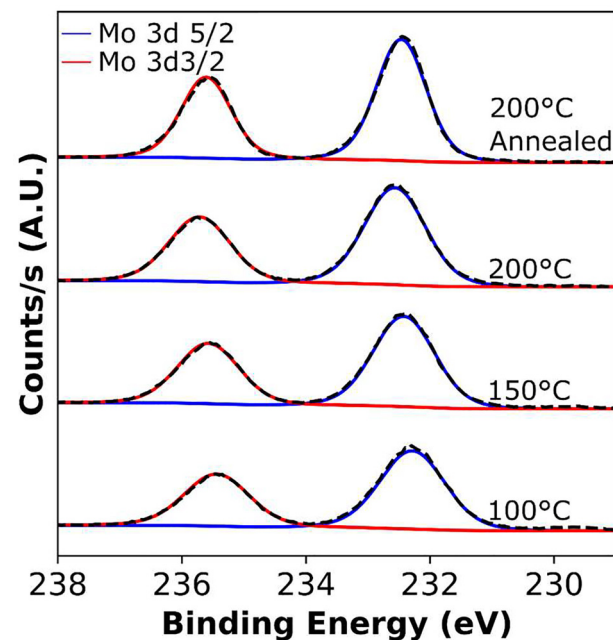


FIG. 4. Mo 3d core-level XPS spectra for XPS ALD MoO_x deposited using $\text{Mo}(\text{PrCp})_2\text{H}_2$ at 100, 150, and 200 °C and a 200 °C deposited film annealed at 350 °C for 20 min.

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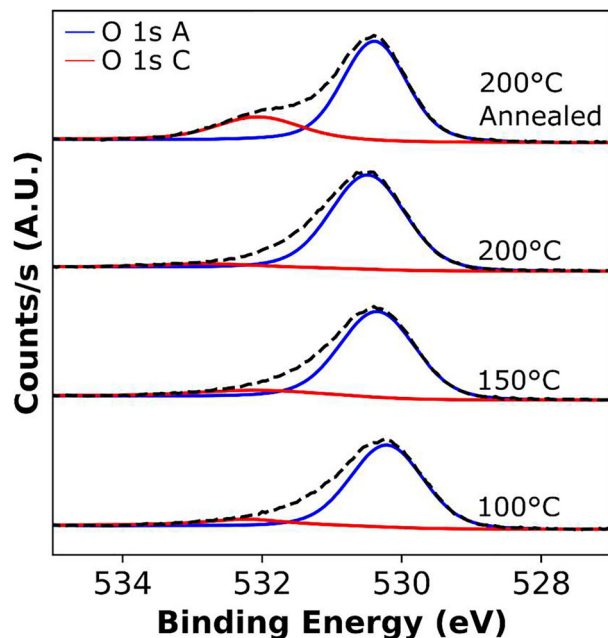


FIG. 5. O 1s core-level XPS data for ALD MoO_x deposited using Mo(PrCp)₂H₂ at 100, 150, and 200 °C and a 200 °C deposited film annealed at 350 °C for 20 min.

at lower temperatures. No trend was observed between roughness and deposition temperature.

The XRD patterns for as-deposited and annealed MoO_x deposited with Mo(PrCp)₂H₂ are shown in Fig. 3. As-deposited films were determined to be amorphous across all deposition temperatures for films deposited with both precursors. After annealing in air for 20 min at 350 °C, a peak arises that is associated with the (040) plane of α -phase MoO₃ ($2\theta \approx 25.5^\circ$),¹⁹ indicating

TABLE III. Elemental concentrations and O:Mo ratio for as-deposited and annealed MoO_x with different precursors. Annealed MoO_x samples are deposited at 200 °C and annealed at 350 °C for 20 min. The carbon concentration is also shown for MoO_x samples after 360 s Ar sputtering.

Mo precursor (deposition temperature and postgrowth processing)	Mo (%)	O (%)	O:Mo	C (%)
Mo(NtBu) ₂ (NMe ₂) ₂ (200 °C)	19.1	50.0	2.62:1	12.0
Mo(ⁱ PrCp) ₂ H ₂ (100 °C)	15.9	52.2	2.78:1	23.9
Mo(ⁱ PrCp) ₂ H ₂ (150 °C)	17.5	55.7	2.63:1	17.1
Mo(ⁱ PrCp) ₂ H ₂ (200 °C)	19.0	63.7	2.75:1	15.9
Mo(ⁱ PrCp) ₂ H ₂ (200 °C and annealed)	17.1	42.2	2.47:1	6.5
Mo(NtBu) ₂ (NMe ₂) ₂ (200 °C and annealed)	23.3	57.0	2.45:1	6.3

crystallization after annealing. No other diffraction peaks were observed for the annealed samples.

The stoichiometry and composition of MoO_x deposited at 100, 150, 200, and 200 °C with postdeposition annealing at 350 °C,

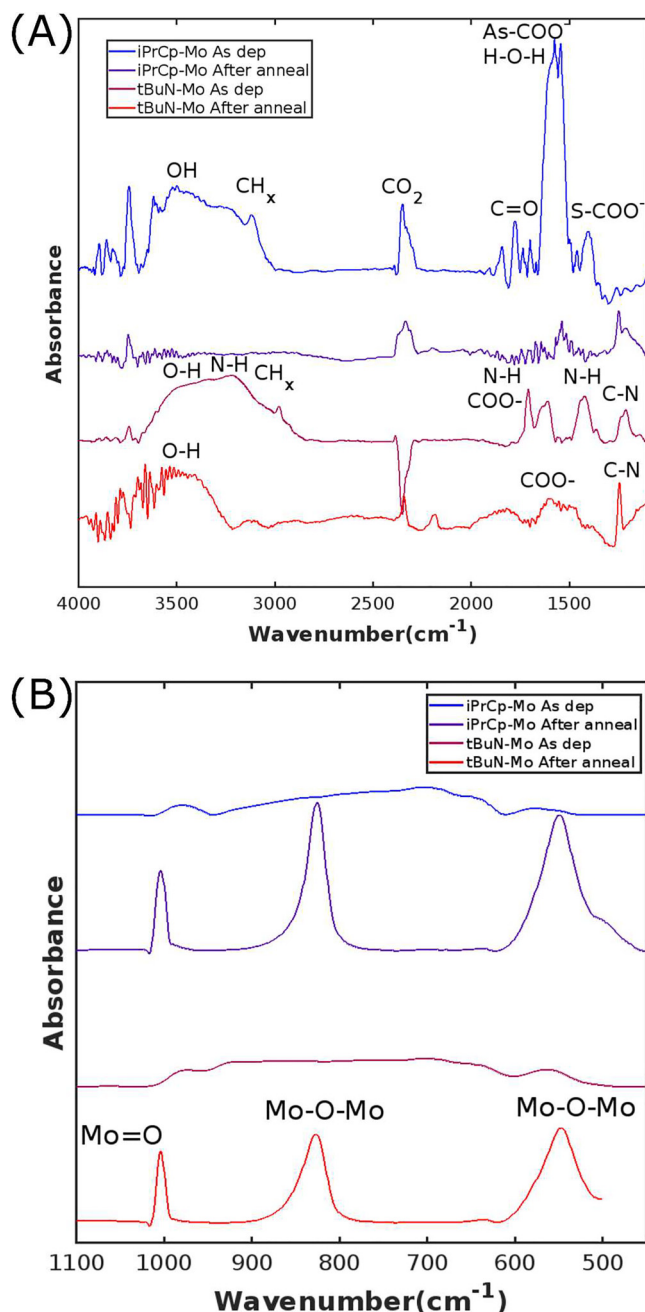


FIG. 6. Ex situ FTIR data of as-deposited and annealed MoO_x film using Mo(PrCp)₂H₂ and Mo(NtBu)₂(NMe₂)₂ from (a) 4000–1100 and (b) 1100–500 cm⁻¹.

were probed using XPS. Among the O 1s and Mo 3d spectra, there were minor shifts in peak binding energies with changes in growth temperature, which may be due to an error in adjusting the data to the adventitious carbon peak at 284.8 eV. Primary components of the Mo 3d_{5/2} and Mo 3d_{3/2} spectra at 232.5 and 235.6 eV relate to the Mo⁶⁺ oxidation state,²⁰ with a concentration of ~18%. The O 1s spectra contain two peaks associated with the Mo—O oxide matrix at 530 eV,²⁰ with a concentration of ~49%. Another peak associated with surface O—H was found at 532 eV.^{18,21,22} Concentration of O—H was under 5% for as-deposited samples, increasing to 11.3% after annealing. The O:Mo ratio was calculated to be slightly less than 3 (Figs. 4 and 5 and Table III).

In the C 1s spectra, intensity decreased by a factor of 1.5–2 for all deposition temperatures after sputtering. However, the concentration of carbon was still larger than typically reported for MoO_x deposited with Mo(NtBu)₂(NMe₂)₂ and may be attributed to differences in O₃ concentration or other process variations. Despite the high carbon concentration in as-deposited samples using Mo(NtBu)₂(NMe₂)₂, films derived from the Mo(ⁱPrCp)₂H₂ precursor contained 11.9%–3.9% more carbon. The high percentage of C implies that Mo(ⁱPrCp)₂H₂ does not fully react with O₃ to remove precursor ligands during deposition. Annealing films deposited with Mo(ⁱPrCp)₂H₂ caused carbon concentration to decrease to 6.5 at. %. However, the reduction of carbon due to annealing does not seem to be affected by precursor chemistry.

FTIR spectra of as-deposited MoO_x films using Mo(ⁱPrCp)₂H₂ and Mo(NtBu)₂(NMe₂)₂ precursors are compared with the FTIR spectra after annealing (Fig. 6). As-deposited films using Mo(ⁱPrCp)₂H₂ contained peaks corresponding to unsaturated aromatic rings (3000–3100 cm⁻¹),^{23,24} triple bonded carbon (3300 cm⁻¹),^{23,24} and asymmetric and symmetric COO⁻ stretch (1650–1540 and 1450–1360 cm⁻¹, respectively).²⁵ The presence of CH_x and COO⁻ species in as-deposited Mo(ⁱPrCp)₂H₂ films is in agreement with the C1s spectra observed in XPS. As-deposited Mo(NtBu)₂(NMe₂)₂ films contained peaks corresponding to N—H amine stretch (3400–3250 cm⁻¹),²⁶ single bond C—H stretching (2850–3300 cm⁻¹),²³ N—H primary bend in amines and asymmetric COO⁻ stretch (1580–1650 cm⁻¹),²⁵ and aliphatic C—N stretch (1020–1250 cm⁻¹).²⁶ For both precursor chemistries, peaks corresponding to O—H (3250–3650 cm⁻¹)²⁶ and CO₂ (2350 cm⁻¹)^{27–29} are observed. The presence of O—H could be attributed to the hygroscopic nature of MoO_x and moisture uptake from the atmosphere. The CO₂ peak is attributed to incomplete removal from the atmosphere after 15 min of N₂ purge. After annealing, peaks corresponding to Mo=O stretching (1003 cm⁻¹) specific to α-phase MoO₃ are observed, as are peaks corresponding to Mo—O—Mo (825, 549 cm⁻¹)^{2,24,30–33} CH_x peaks are lost in both annealed films, indicating the removal of hydrocarbon species after annealing (Fig. 6).

IV. SUMMARY AND CONCLUSIONS

Substoichiometric MoO_x with significant C content was grown using atomic layer deposition in the temperature range of 100–225 °C using Mo(ⁱPrCp)₂H₂ and O₃. GPC in the ALD window ranged from 1.3 to 1.7 Å/cycle. Films are amorphous as-deposited and crystallized to α-MoO₃ after annealing at 350 °C in air.

Compositional analysis revealed high carbon concentrations in as-deposited films that were slightly higher relative to the comparison amido Mo precursor, with these concentrations decreasing after annealing. The C concentration might be reduced by using a higher concentration of ozone or an oxygen plasma. In comparison to films deposited using Mo(NtBu)₂(NMe₂)₂ and O₃, films grown with Mo(ⁱPrCp)₂H₂ have a higher GPC and lower density.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Ethan Hendrix: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Benjamin Garland:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Writing – review & editing (equal). **Vamseedhara Vemuri:** Data curation (equal); Formal analysis (equal); Methodology (equal); Writing – original draft (equal); Writing – review & editing (equal). **Nicholas C. Strandwitz:** Conceptualization (equal); Data curation (equal); Funding acquisition (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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